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Review article

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Towards system-level artificial intelligence in perovskite photovoltaics

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Abstract

Halide perovskite solar cells (PSCs) offer tunable optoelectronic properties and low fabrication costs; however, their industrial deployment is constrained by tightly coupled dependencies among composition, processing and device architecture. Artificial intelligence (AI) has been widely applied across the PSC research cycle, but its impact remains uneven: models accelerate local optimization within data-rich regimes yet do not generalize across compositions, fabrication environments or device architectures. This unevenness reflects the difficulty of representing the coupled, cross-scale relationships that govern PSC behaviour rather than a limitation of modelling capability alone. We examine AI applications across materials discovery, device engineering, process optimization and stability analysis, identify four structural bottlenecks (fragmented data, weak generalization, limited interpretability and misaligned design objectives) and outline research directions to address each. Broad impact will require a transition from task-specific predictive tools towards system-level AI integration that preserves knowledge across the PSC research cycle.

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Key points

- Perovskite solar cell (PSC) deployment is bottlenecked by tightly coupled dependencies, not by single-parameter optimization. Composition governs processing dynamics, processing determines microstructure, and microstructure sets device performance, so improvements identified in one system rarely transfer to another.
- Artificial intelligence (AI) accelerates PSC research locally but does not generalize. Across materials discovery, device engineering, process optimization and stability analysis, AI reliably identifies correlations and speeds up optimization within data-rich regimes; it does not yet extract mechanism-anchored insights that transfer across compositions, architectures or operating conditions.
- The limitation is structural, not algorithmic. Four bottlenecks recur across AI applications in PSCs: fragmented and non-interoperable data, weak generalization across systems, limited interpretability with weak physical grounding, and misalignment between AI objectives and practical design constraints.
- Addressing these bottlenecks requires integrated infrastructures, not more accurate individual models. Multimodal and multilevel datasets, physics-informed and explainable models and closed-loop coupling between AI predictions and experimental feedback each address one bottleneck; together, they provide the building blocks for a coupled research pipeline.
- System-level AI integration is the direction of travel. The near-term shift is from local, task-specific AI towards frameworks in which knowledge is continuously propagated, updated and reused across the full PSC research cycle, reframing AI from a predictive tool into a participant in materials discovery, device optimization and reliability engineering.

Introduction

Halide perovskite solar cells (PSCs) have reached certified power conversion efficiencies (PCEs)[1] of 28%, and their tunable optoelectronic properties, solution processability and low fabrication costs[2] have made them one of the most active platforms in photovoltaic research. However, the path to industrial deployment remains constrained by instability under heat, moisture and illumination, by the presence of lead-based components and by a tightly coupled interplay among material chemistry, processing conditions and device architecture[3–5]. These couplings mean that improvements identified in one composition, process or architecture rarely transfer to another.

Artificial intelligence (AI), spanning machine learning (ML), deep learning and data-driven optimization, has therefore been applied across the PSC research cycle, from bandgap E_g prediction and materials screening to process optimization and device engineering[6,7]. Within well-defined systems and data-rich regimes, these approaches deliver measurable acceleration: they identify correlations that guide experimental design, narrow candidate compositions and rank the variables that most strongly influence performance.

Progress in these areas has, nonetheless, been uneven. Most existing efforts remain narrowly scoped, relying on isolated predictive models with limited generalizability, weak links across composition, processing and performance and minimal coupling to real-time experimental feedback[8].

The underlying difficulty is not primarily algorithmic. It reflects the difficulty of capturing the coupled and evolving relationships among chemistry, processing, device operation and degradation. These relationships are particularly demanding in PSCs, in which ion migration, crystallization dynamics, interfacial reactions and metastability generate strong cross-scale dependencies that are difficult to capture using conventional data-driven models.

We refer to the research paradigm needed to address these dependencies as system-level AI integration: a mode of working in which knowledge generated at one stage of the PSC research cycle is continuously propagated, updated and reused to inform subsequent stages through coupled digital–physical feedback. What distinguishes it from conventional closed-loop optimization, which improves individual tasks in isolation, is this continuous cross-stage propagation of transferable knowledge. Workflows that optimize only a single task without preserving and propagating information beyond it fall short of this definition.

This Review develops that argument in three parts. We first survey what AI has and has not delivered across materials discovery, device engineering, process optimization and stability analysis in PSCs, drawing out a shared pattern of local acceleration but weak transferability. We then examine the four structural bottlenecks that limit broader impact (fragmented data, weak generalization, limited interpretability, and misalignment with practical design objectives) and the research directions that could address each. We close by sketching a route from today's isolated workflows towards system-level AI integration across the PSC research life cycle.

AI capabilities and limits

AI has been applied at every stage of the PSC research cycle to address the challenges outlined earlier: materials discovery, device engineering, process optimization and stability analysis (Fig. 1). Across these stages, common strengths and limitations emerge. AI reliably identifies patterns within well-defined systems and accelerates local optimization, but it extrapolates poorly beyond narrow compositional and processing windows, offers weak transferability across device architectures and provides limited mechanistic insight. Considered together, these stage-specific limitations point to a shared cause: the fragmentation of AI workflows across an otherwise tightly coupled research pipeline. This fragmentation frames the case for system-level integration developed later in this Review.

Materials discovery and extrapolation

AI has proved most useful in compositional exploration. For properties such as the bandgap E_g, in which large and internally consistent datasets exist, models can map composition–property relationships more efficiently than experiment permits[9–11]. These models enable navigation of multicomponent systems, including mixed-cation and mixed-halide compositions, without relying entirely on trial and error.

This approach has been extended to less conventional systems, including low-dimensional variants, structures modified by ligands and Pb-free alternatives[7,12–15]. AI is often used to relate molecular characteristics such as ligand size or polarity to structural outcomes such as octahedral distortion, layer spacing or phase stability[16–18]. These relationships are meaningful and, in some cases, provide useful guidance for narrowing candidate materials.

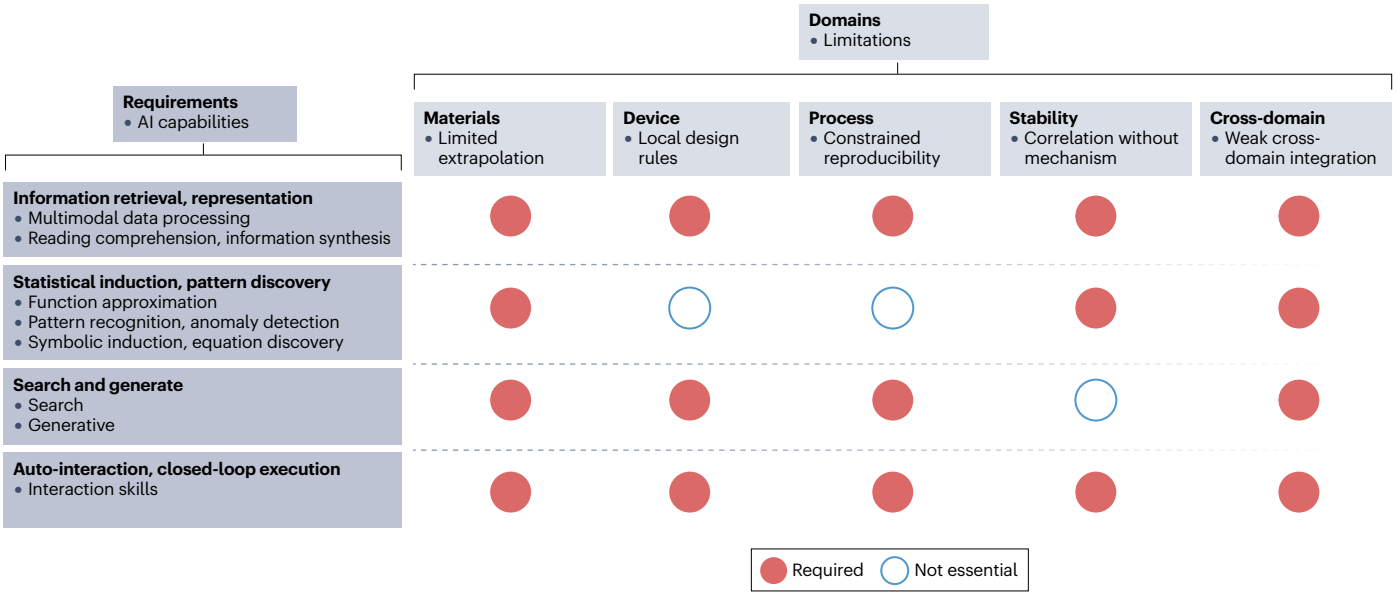


Fig. 1 | Capabilities and limitations of artificial intelligence across the perovskite solar cell research workflow. Matrix of AI (artificial intelligence) capabilities required to address the principal limitation in each domain of perovskite solar cell research. Filled circles indicate a required capability; open circles indicate one that is not essential.

Predictive performance, nonetheless, deteriorates rapidly once the chemistry moves outside the compositional range of the training data[19]. Unusual ligands, new B-site substitutions or competing phase behaviour introduces effects (defect formation, lattice instability and phase segregation) that are only weakly represented in existing datasets[14,15]. As a result, models extrapolate poorly and remain better at organizing known compositions than at identifying genuinely new materials. Most studies accordingly remain retrospective analyses of published data, with few examples in which AI predictions have guided experimental discovery of a new composition.

Device engineering and local design rules

At the device level, AI has helped identify the factors that most influence device performance, including energy level alignment, interfacial recombination and charge transport balance[20,21]. This has enabled more efficient screening of transport layers, interface modifiers and additive systems, particularly in architectures in which multiple variables interact nonlinearly[22–24].

In practice, AI often acts as a filter, narrowing candidate materials and highlighting design directions, for example, by identifying interfacial treatments that suppress non-radiative recombination or additives that reduce defect density at grain boundaries[24–26]. These insights can accelerate optimization within specific device architectures.

However, these findings rarely transfer across systems. Small variations in composition, interface chemistry or processing conditions can alter device behaviour, so that the predictive power of models trained on one dataset does not carry over to another. Correlations identified at the device level do not necessarily reflect underlying mechanisms such as defect passivation pathways or carrier recombination dynamics. AI, therefore, supports local optimization but does not provide the mechanism-anchored design principles that would generalize across device platforms.

Process optimization and narrow optima

If there is a domain in which AI delivers immediate practical value, it is process optimization[27]. Fabrication of PSCs involves many coupled parameters, such as solvent choice, precursor ratios and annealing conditions, and exploring this space experimentally is time-consuming[28]. AI can shorten this process by steering experiments towards regions likely to yield high performance[29–33].

Processing cannot, however, be reduced to a set of static parameters. Film formation depends on dynamic phenomena: nucleation behaviour, intermediate phase formation, solvent evaporation rates and grain growth pathways. The incorporation of in situ diagnostics has made this dependence explicit, revealing how small variations in these dynamics can lead to differences in morphology and device performance[34,35].

The ‘optimal’ conditions identified by AI are often confined to a narrow window, so many AI-selected solutions are not robustly optimal but rather context-sensitive local optima. This narrowness helps explain why reproducibility across laboratories remains difficult and why scale-up to manufacturing continues to lag laboratory optimization.

Stability without mechanistic insight

Stability is arguably the most complex aspect of PSC research, and AI has been widely applied to it. By analysing time-dependent data such as photoluminescence evolution or spectral changes under stress, AI can identify which variables are most strongly associated with degradation[36–38].

These analyses help spot early warning signs: phase segregation under illumination, the onset of ion migration or the accumulation of defects that eventually lead to performance loss[10,38]. They also permit comparisons across different stress conditions, helping to rank the relative importance of heat, moisture and electric fields[39–42].

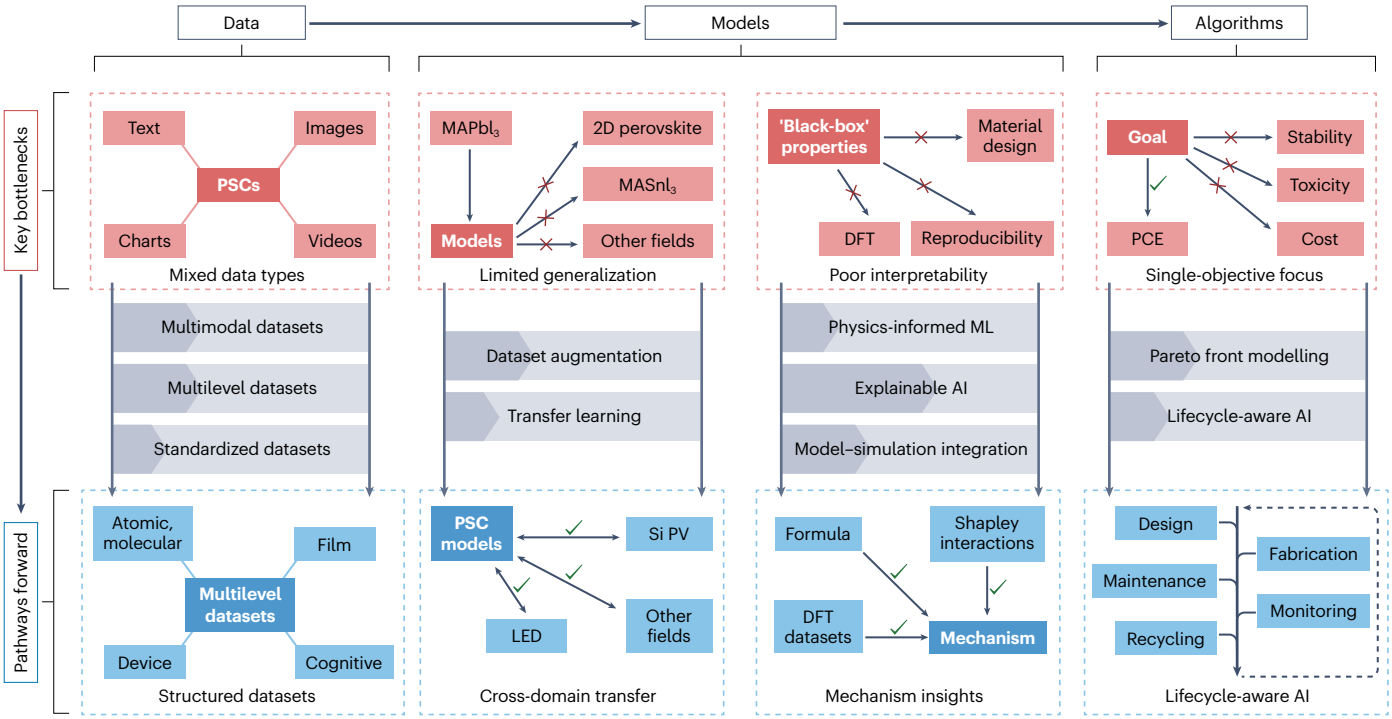


Fig. 2 | Structural bottlenecks limiting the practical impact of AI. Four bottleneck themes (red panels) are arranged along a data–models–algorithms axis: mixed data types, limited generalization, poor interpretability of black-box models and single-objective focus. Grey bars list the strategies proposed to address each bottleneck, and blue panels show the pathway outcomes each strategy targets. The vertical arrow at the left marks the progression from present bottlenecks to future pathways. DFT, density functional theory; LED, light-emitting diode; MA, methylammonium; ML, machine learning; PCE, power conversion efficiency; PSC, perovskite solar cell; PV, photovoltaics.

Most such analyses, nonetheless, operate at the level of correlation. They identify which factors matter, but often do not capture how degradation pathways unfold at the microscopic level, for example, how ion migration interacts with defect formation or phase transitions over time. Combined with the absence of standardized long-term datasets, this limitation makes reliable lifetime prediction difficult. For now, AI organizes stability data better than it fully explains them, and generalizable lifetime prediction remains an open challenge.

Cross-domain limits and weak integration

Across materials discovery, device engineering, process optimization and stability analysis, a consistent limitation is evident: AI excels at high-dimensional correlation analysis and local optimization but does not deliver system-level, generalizable insights[43]. This is not merely a model shortcoming but a consequence of the tightly coupled nature of PSC systems, in which composition governs processing dynamics, processing defines microstructure, and microstructure determines device performance. Isolated treatment of these interconnected stages inevitably discards the mechanistic couplings that link them.

PSCs pose an unusually difficult challenge for AI because device behaviour emerges from tightly coupled and dynamically evolving processes that span multiple spatial and temporal scales. Ion migration continuously modifies local electric fields and defect distributions during operation, whereas crystallization pathways remain highly sensitive to environmental fluctuations and processing history. Metastable phases, interfacial reactions and stress-induced structural evolution introduce non-equilibrium behaviour that is difficult to represent using static descriptors. As a result, correlations learnt from one dataset often become invalid when transferred across compositions, architectures or operating conditions.

Viewed from this perspective, the challenge is not simply to boost predictive accuracy, but to reframe research workflows around integrated, cross-scale frameworks that unify data, models and experiments. Such a paradigm shift will be as consequential for unlocking AI's potential in PSC engineering as incremental algorithmic improvements.

Structural bottlenecks for AI

Reliable, widespread deployment of AI in PSC research is constrained by four interrelated bottlenecks (Fig. 2): fragmented data with weak interoperability, limited generalization across materials and processes, weak physical grounding of black-box models and misalignment between AI objectives and practical design constraints. Each is examined below alongside the research directions that could shift AI from a correlational tool towards a mechanistically grounded design partner.

Data gaps and interoperability limits

PSC research suffers from fragmented datasets with inconsistent formats, uncontrolled variability and limited public accessibility. Experimental data are often recorded in unstructured formats such as figures and supplementary files and lack essential metadata such as processing conditions, measurement protocols or degradation history. Three related deficits, therefore, need to be addressed: the difficulty of

jointly using different data modalities, the absence of representations that span the atomic to the device scale and the lack of standardized, dynamically maintained datasets.

Multimodal data integration challenges. PSC data are multimodal. Research should therefore not rely on simple correlation between literature-mined descriptors and performance. A well-formed dataset should integrate explicit data modalities (morphology images, spectroscopy, current–voltage and external quantum efficiency curves and ageing profiles) with structured metadata (fabrication parameters and scalar electrical metrics), so that models can learn the relationships among composition, process and performance[44]. Monomodal data often capture only a subset of the relationships governing PSC behaviour and does not reflect fundamental physical principles. For example, materials with identical compositions can differ in morphology and stability because of varying processing conditions[45–47]. Therefore, reliable predictions of performance and stability require joint modelling that combines composition, morphology, spectroscopic and electrical characterization.

We distinguish multimodal data from multisource and multilevel data. Multimodal data are fundamentally different forms of information, such as morphology images, spectroscopy, electrical characterization curves and ageing trajectories. By contrast, structured metadata provide contextual information about how each data record was obtained. Multisource and multilevel data capture information across different experimental pathways and physical scales. Effective PSC learning ultimately requires the integration of all three dimensions across composition, processing and device operation.

Multilevel data representation gaps. PSC research requires the construction of rigorous, multilevel datasets to bridge computational and experimental feasibility[48,49]. Such datasets span four levels: atomic and molecular, thin-film, device and (as elaborated below) a knowledge-level representation that captures cross-level relationships. Models trained solely on data from a single level tend to produce isolated black-box mappings, leading to mismatches between materials and experimental or device results[50–52]. By contrast, multilevel training incorporates cross-level physical constraints, which improves prediction accuracy and reveals mechanistic evolution[44,53]. For example, correlating the spectral signals of perovskites with morphological heterogeneity can pinpoint the microscopic origins of device degradation[54,55].

Standardization and dynamic updating. A further data challenge is insufficient standardization and poor interoperability. In many subfields, chemical composition records lack a standardized format, making them difficult to convert into uniform atomic descriptors[56]. Furthermore, multimodal data are often compressed into scalar values, resulting in the loss of information[54,57]. Data integrity is compromised by sparse structures and inconsistent handling of missing values; in some databases, mixed notation and ambiguous field definitions can lead to ambiguity[58]. From a distributional perspective, literature-mining databases also suffer from positive bias and long-tail distribution issues — for example, an excess of data on TiO_2 electron transport layers, which can lead models to learn editorial preferences of the field rather than universal physical laws. Finally, static dataset construction cannot keep pace with the rapid evolution of PSC technology[56]. Without dynamic update mechanisms, feature spaces lag behind current practice in the field.

Although the integrated multimodal, multilevel and standardized database envisaged earlier remains conceptual, the continued emergence of monomodal databases offers a foundation on which such a resource can be built; concrete recommendations are set out in the Outlook section.

Weak generalization across systems

Most published models are system-specific and trained on narrow datasets, for example, laboratory-scale $MAPbI_3$-based systems, resulting in weak generalization. These models cannot easily accommodate variations in dimensionality, chemical system and fabrication process. The challenge is not simply one of data quantity but the difficulty of preserving physicochemical invariants across heterogeneous PSC systems. Data augmentation and transfer learning should therefore be viewed not as standalone solutions but as attempts to recover transferable representations from incomplete observations.

Dataset augmentation limitations. In perovskite research, sample scarcity makes it difficult to construct standardized databases, and this scarcity can lead to model underfitting and reduced prediction accuracy. To address this scarcity, common data-level augmentation strategies used in perovskite research include conventional data transformation, generative adversarial networks and large-language-model-based data generation technologies[59,60]. For example, elemental cation substitution at the A-site and B-site of perovskite lattices is widely used to expand small original datasets and diversify sample coverage[61].

Data augmentation has successfully increased dataset sizes. However, increasing dataset size does not necessarily improve dataset quality[62]. A challenge is that generated samples may satisfy statistical similarity to real data while violating fundamental physicochemical constraints. For PSC applications, charge neutrality, phase stability, tolerance factor requirements, synthetic accessibility and device relevance should be considered during both data generation and validation (Table 1). Without such constraints, augmented datasets may introduce

Table 1 | Data augmentation for different data types

Data type	Use case	Augmentation	Constraint	Validation
Composition	Materials screening	Constrained substitution; generative models	Charge neutrality; phase stability	Physical feasibility
Process	Process optimization	Parameter perturbation; generative models	Process window	Experimental validation
Images	Morphology analysis	Rotation; crop; mixup	Morphology preservation	Visual consistency
Spectra	Phase and defect analysis	Noise; shift; scaling	Peak assignment	Spectral consistency
Device ageing	Stability prediction	Resampling; physics-based simulation	Ageing trend	Lifetime-test consistency

chemically unrealistic compositions or experimentally inaccessible structures, thereby limiting model reliability and transferability.

Transfer learning and domain mismatch. Although data augmentation acts within a single system, transfer learning, which reuses a model trained on one system as the starting point for a related system, transfers general knowledge, structural patterns and features from the source domain to a different but related target domain. By using general structure–property relationships, transfer learning reduces the computational and time costs of training from scratch.

However, transfer learning has yielded limited results in perovskite materials research so far and can suffer from negative transfer caused by cross-domain heterogeneity[63]. In a typical failure case, models trained on the dataset of lead–halide 2D perovskites processed by conventional solution methods produce unreliable predictions when applied directly to Sn-based or Bi-based perovskite systems[61]. The distinct ionic radii, valence electron configurations and lattice-distortion characteristics of the two families break the shared structure–property correlation the model has learnt[64]. Similarly, models optimized for 2D perovskite architectures often do not transfer to other architectures, because charge transport requirements and bonding mechanisms vary across device structures[65].

More fundamentally, transfer failure often arises because models learn dataset-specific statistical regularities rather than invariant physicochemical relationships. When the dominant degradation mechanisms, crystallization pathways, defect chemistries or interfacial interactions differ between source and target domains, correlations learnt from one system become unreliable predictors in another. As a result, models that perform well within a narrowly defined materials family or processing protocol frequently fail when transferred across compositions, device architectures or operating conditions. This challenge is particularly severe in PSC research because device performance emerges from coupled interactions among composition, processing history, microstructure and operational conditions, many of which are poorly represented in existing datasets. A shared data space must therefore be carefully designed to avoid mismatches in structure and features and to eliminate systematic experimental biases. The precursor space produced by transfer learning still requires validation by small-scale, high-throughput experimentation to quantify gains in prediction accuracy.

Interpretability and physical grounding

A criticism of AI in materials science is its black-box nature. High-performing models often provide little insight into why a given prediction is made. In the PSC context, this both constrains mechanism-guided design of materials and devices and impedes the integration of ML with density functional theory (DFT), device modelling and process simulation, and it reduces the reliability of research results and inter-laboratory reproducibility. Three responses to these limitations are considered below: physics-informed ML (PIML), which embeds known physical laws into the model's architecture or training objective; explainable AI, which exposes which input features drive a model's predictions; and the joint use of models with physical simulation.

Physics-informed modelling strategies. In PSC research, the main barrier to interpretable AI is that purely data-driven models cannot account for the strong nonlinear couplings among the dynamic processes that occur inside a PSC. PIML is particularly suited to PSC systems, because it can embed lattice dynamics, carrier transport, ion migration and defect physics into neural networks, so that models can infer causal relationships among processing history, microstructure evolution, device operation and long-term stability, rather than relying on empirical correlations.

However, multiscale physical coupling remains conceptual in practice. This is because many key phenomena in perovskites evolve across multiple spatial and temporal scales and cannot be fully characterized by a single physical constraint or statistical correlation. For example, ion migration continuously modifies local electric fields and interfacial energetics during device operation, whereas crystallization pathways determine defect distributions and microstructure formation during film growth. In addition, degradation processes emerge from the coupled evolution of defects, strain, phase segregation and interfacial reactions under external stresses.

Phase transitions, halide segregation, lattice reconstruction and black-phase degradation are rare events involving bond-breaking and restructuring, so that the physical loss function of PIML cannot precisely constrain the kinetic pathways. Organic–inorganic hybrid perovskites also involve covalent, ionic and van der Waals interactions, for which no unified analytical physical prior is available.

Explainability and its limitations. A range of explainable AI techniques, such as Shapley additive explanations, have been used to analyse feature importance and decision logic, thereby increasing human trust in models[64]. Although these methods, and their combinations with other approaches, appear to provide interpretable insights for optimizing PSC performance and exploring underlying mechanisms, long-standing puzzles remain unresolved[20,37,65].

Existing mechanistic studies have largely focused on rationalizing experimental observations within established frameworks, whereas fundamentally new mechanistic perspectives remain scarce. For example, a particular halide ion has been reported to dominate both the bandgap[66] and the perovskite adsorption energy[67]. This convergence on a single feature may reflect subjective feature selection or the omission of features judged unimportant a priori. Future opportunities may therefore lie in applying explainable AI to phenomena that remain difficult to probe experimentally, including metastable ion migration pathways, high-temperature degradation and crystallization dynamics, in which AI may help reveal hidden variables and causal relationships[39].

Furthermore, biases arising from correlations between element features should also be reduced[68]. Explainable AI should not be limited to explaining 'material design'; explaining 'model design' will improve the accuracy of model predictions.

Model and simulation integration. Integrating ML models with DFT and time-dependent DFT simulations has become a standard workflow in perovskite calculations[69,70]. Typically, molecular screening from source databases to intermediate or synthetic databases relies on random selection or feature engineering to identify meaningful descriptors[22,71]; however, this approach remains constrained by the subjective judgement of the researcher. By contrast, molecular descriptors derived from DFT calculations in intermediate databases shift the focus from human-defined macroscopic features to microscopic quantities.

Automating the evaluation of DFT convergence, the selection of functionals and the diagnosis of strong correlations could reduce human intervention in ML and may become a core component of autonomous design. Although cross-validating DFT results against

ML predictions can improve model accuracy, the scarcity of standardized DFT datasets, together with the inherent systematic errors and the supercomputing cost of high-precision DFT data, undermines the efficiency motivation for using ML in the first place.

Misalignment with design objectives

Many AI models in PSC research target only maximizing PCE. However, industrial feasibility requires balancing PCE against long-term operational stability, toxicity and environmental impact, and scalability against material costs and supply-chain constraints. Multi-objective optimization (MOO) frameworks that map trade-offs on a Pareto front, together with lifecycle-aware AI, are needed to move PSCs from the laboratory to commercialization.

Multi-objective optimization frameworks. MOO has been widely applied in perovskite research because it addresses problems that require coordinated screening of multiple metrics, including device design, interface control, process sensitivity and performance coupling. It reduces experimental and simulation costs while streamlining optimization from material screening to device fabrication[72]. Combined with high-throughput experiments and ML-assisted experimental design, MOO supports systematic frameworks for identifying component and doping schemes that balance PCE and long-term stability[73].

However, MOO's decision-making depends on human-defined objectives and offers limited insight into the structure–property mechanisms of perovskites. Furthermore, MOO may remain confined to laboratory research, as there is no unified quantitative standard for weighting multiple objectives in industrial applications.

Lifecycle-aware design strategies. When assessing the market potential and industrialization progress of PSCs, it is essential to consider performance, stability, cost, environmental impact and device lifespan. AI that incorporates lifecycle metrics enables sustainability assessment across PCE, toxicity and environmental burden[43]. For example, the green solvents γ-pentyl acetate and ethyl acetate can reduce the safety risks and costs of traditional solvent systems while maintaining a high PCE[43,74]. From a full-lifecycle perspective, PSCs based on such solvent systems can become competitive with Si photovoltaics and other renewable energy sources.

Although lifecycle-aware AI remains conceptual in the perovskite field, it is central to moving from the laboratory to commercialization[75]. Its development requires the synergistic advancement of data, models and MOO, together with a unified digital foundation and industry standards spanning the full chain from research and development to recycling. It must also integrate cross-scale physical mechanisms to achieve closed-loop management across the full life cycle.

These four bottlenecks — fragmented data, weak generalization, limited interpretability and misaligned objectives — are interconnected rather than independent. Each ultimately reflects the same underlying challenge: current AI workflows cannot represent or learn the coupled processes that span composition, processing, device physics and degradation across scales. This shared cause motivates the integrated AI frameworks discussed next.

Outlook

It is tempting to attribute progress in AI for PSC research to increasingly powerful models. In practice, however, many of the limitations discussed earlier do not arise from modelling capability alone; they reflect a mismatch between how data are used, how models are trained and how experiments are carried out. The future of AI in PSC research will therefore be shaped less by incremental algorithmic improvement than by frameworks that integrate data, physics, experimentation and operational feedback.

Progress will therefore depend not only on model architecture but also on the quality, diversity and interoperability of experimental datasets and on knowledge transfer across all PSC subtasks in both the physical and digital realms.

Dynamic and physics-aware modelling

Processing in PSC systems is inherently dynamic; however, it is often treated as a static optimization problem. In reality, crystallization pathways, intermediate phases and solvent dynamics evolve over time and are highly sensitive to environmental conditions[6]. Small fluctuations in humidity or temperature can alter nucleation behaviour and grain growth, producing different defect distributions and device performance[76]. Such effects are difficult to capture with models that rely only on fixed input parameters.

Moving forward, incorporating time-dependent and environment-sensitive information will be essential. This may require the integration of real-time diagnostics or treating processing as a trajectory rather than a set of discrete conditions. The challenge is not only to predict outcomes but also to capture how those outcomes emerge during film formation.

Multimodal data infrastructures

PSC datasets must go beyond simple correlations between literature-mined descriptors and performance. They must integrate multiple data modalities and structured metadata so that models can consume multisource data and link information across scales

Table 2 | Development of AI capabilities at different stages

	Design	Forecast	Knowledge	Analysis	Fabrication	Test	Preparation
Data-driven prediction	Manual	PCE forecast	Curated data	Fundamental	Manual synthesis	Manual characterization	Manual configuration
AI-guided design	AI-suggested	PCE forecast	Database	Fundamental	Manual synthesis	Manual characterization	Manual configuration
Local closed-loop workflows	Self-designed	Multi-objective	Auto-updated	Mechanistic	High-throughput	High-throughput	High-throughput
System-level autonomous framework[a]	Self-designed	Multi-objective	Auto-updated	Mechanistic	High-throughput	High-throughput	High-throughput

AI, artificial intelligence; PCE, power conversion efficiency. [a]In the system-level autonomous framework, the digital loop is closed and AI can iterate between the digital and physical loops.

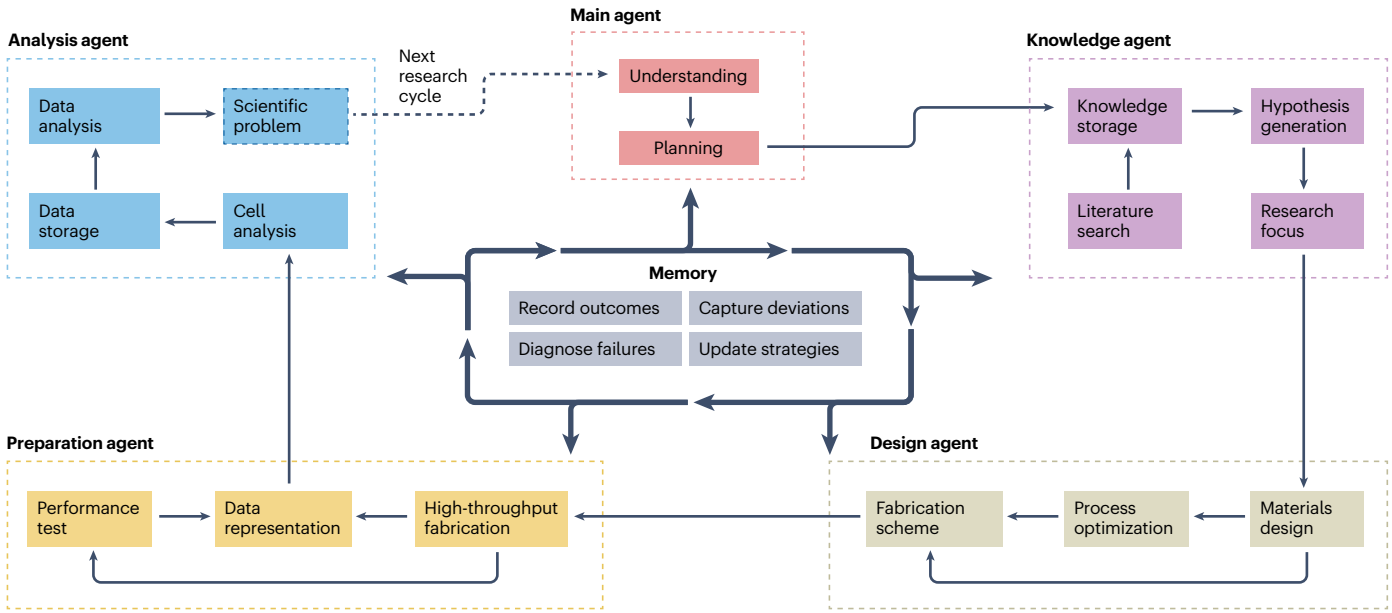


Fig. 3 | Closed-loop AI–experiment integration workflow for PSC development. The schematic illustrates a future system-level AI framework in which knowledge generation, experimental execution and model updating are continuously coupled through iterative feedback loops across the PSC research cycle. Cycles begin at the Main agent and return to it after Analysis, initiating the next research cycle. AI, artificial intelligence; PSC, perovskite solar cell.

and dimensions. This requires structured, detailed and standardized archiving of material composition, fabrication processes, device structures and testing conditions. The archive must also support compliant uploads and cross-verification by research teams worldwide, so that the resulting public dataset is reusable, reproducible and comparable.

The archive should draw on established industry standards such as the National Renewable Energy Laboratory Photovoltaic Data Acquisition (NREL-PVDAQ) database for the photovoltaic sector and on authoritative computational and experimental databases such as the Inorganic Crystal Structure Database (ICSD) and Materials Project for inorganic crystalline materials. It should incorporate modules for standardized feature extraction, data normalization and physical-constraint validation, so that it aligns with ML modelling requirements and integrates with explainable AI and physics-informed learning frameworks. Such an archive would support an integrated research-and-development platform for perovskite material screening, device optimization, mechanism analysis and industrial parameter iteration.

Closed-loop AI–experimental integration

A related direction is the integration of AI with experimental workflows[77]. Rather than treating prediction and validation as separate steps, these approaches create iterative cycles in which models guide experiments and are continuously updated by new data.

In PSC research, such closed-loop workflows could bridge stages that are currently separated: for example, composition design, processing optimization and device evaluation[78]. Implementing them remains challenging. Experimental platforms need to execute precisely and provide reliable data, whereas feedback models must adapt as new information becomes available. At least in the near term, the goal is less about full automation than about strengthening the feedback loop between decision-making and physical validation.

Towards system-level AI integration

To realize the potential of AI in PSC research, the field must move beyond isolated prediction tasks towards integrated, iterative workflows that link data maturity, model development and experimental validation into a continuous feedback system. Here, we outline a two-part roadmap: first, a conceptual four-stage progression of AI-enabled research paradigms (Table 2), and second, the transition from local closed-loop workflows to a global system-level framework (Fig. 3).

Four-stage evolution of AI paradigms. The progression outlined subsequently reflects a progressive increase in the coupling among data, models and experiments within PSC research. The progression is conceptual rather than chronological and represents the expanding role of AI from retrospective analysis to increasingly active participation in experimental decision-making.

At present, most work remains focused on data-driven prediction. Models are trained on curated datasets to estimate device performance or material properties, often achieving good retrospective accuracy within well-defined systems. However, these approaches remain limited by the scope and quality of available data.

A second stage involves design-oriented tasks, in which AI is used not only to predict outcomes but also to suggest candidate materials or processing conditions. Here, the focus begins to shift towards incorporating constraints such as stability, toxicity or manufacturability, beyond purely performance-driven optimization. However, candidate validation still relies largely on human intervention.

A third stage introduces local closed-loop workflows, either through robotic experimentation or through digital multi-agent systems[38,79–81]. Although these approaches shorten the distance between decision-making and validation, they remain confined to individual tasks or platforms[82]. Looking ahead, a global autonomous PSC framework would generate, test and refine hypotheses continuously

Table 3 | Local and global closed-loop frameworks compared

Framework	Feedback	Scope	Data	Coupling	Knowledge transfer	Maturity
Robotic experimentation	Physical	Single task	Process records	Tight	Limited	Established
Closed-loop optimization	Digital	Task-specific	Structured datasets	Partial	Local	Demonstrated
Self-driving laboratory	Mixed	Platform	Machine-readable	Closed loop	Intra-platform	Emerging
System-level AI	Digital–physical	Full life cycle	Multimodal, multilevel	Cross-stage	Continuous	Conceptual

by treating experimental outcomes as inputs for ongoing model improvement.

From local to global integration. Existing autonomous AI paradigms differ primarily in the scope of knowledge propagation: robotic experimentation, closed-loop optimization and self-driving laboratories remain largely confined to individual tasks or platforms, whereas system-level AI integration enables continuous knowledge transfer across the entire PSC research life cycle.

The motivation for such integration is strong in PSC systems, in which performance emerges from tightly coupled interactions among composition, processing and device architecture[83]. At present, most local closed-loop AI–experiment workflows are closed around a defined task, objective, platform or protocol. Unlike local closed-loop optimization, PSC research requires AI to preserve coupled relationships across materials chemistry, processing, interfaces, device physics and stability. A system-level framework therefore links hypothesis generation, experimentation and model updating through continuous cross-stage knowledge propagation[84]: models propose candidate materials or processing strategies, experiments evaluate their performance under controlled conditions and the resulting data are used to refine subsequent predictions. Over successive iterations, this process can improve both predictive accuracy and experimental efficiency. The challenge is therefore not to close a simple loop but to determine how much of the PSC research workflow the loop can meaningfully connect.

Glossary

2D perovskite
A layered variant in which insulating organic spacer molecules separate perovskite octahedral slabs, giving increased moisture tolerance but reduced charge transport.

A-site and B-site
The two cation positions in the ABX_2 perovskite lattice: the A-site hosts a large cation and the B-site a smaller cation.

Bandgap
The minimum energy required to promote an electron from a semiconductor's valence band to its conduction band. In perovskites, it determines which wavelengths of sunlight are absorbed.

Black-phase degradation
Loss of the photoactive black perovskite phase through conversion to a non-photoactive polymorph.

Crystallization pathway
The sequence of nucleation, intermediate-phase formation and grain growth by which a perovskite thin film transitions from precursor solution to solid film.

Electron transport layer
Thin layers adjacent to the perovskite absorber that selectively extract electrons and block the opposite carrier.

Generative adversarial network
A pair of neural networks (generator and discriminator) trained against each other to produce synthetic data samples resembling a target distribution.

High-throughput experimentation
Parallelized experimental workflows in which many samples are prepared and characterized in an automated pipeline.

Ligand
An organic molecule bonded to the perovskite lattice or a colloidal nanocrystal surface, used to tune structural, electronic or interfacial properties.

Metastable phase
A structural configuration of the perovskite that is not the thermodynamic ground state but is kinetically trapped and can persist under operating conditions.

Multi-agent system
A software framework in which multiple autonomous AI agents cooperate or negotiate to perform tasks that a single agent cannot easily handle.

Non-radiative recombination
Loss of a photogenerated electron–hole pair through processes that do not emit a photon; a primary limit on solar cell voltage.

Octahedral distortion
Deviation of the BX_6 coordination octahedron from ideal geometry, which shifts electronic structure and phase stability.

Pareto front
The set of solutions to a multi-objective problem in which no objective can be improved without worsening another.

Passivation
Surface or interface treatment that neutralizes defects capable of trapping charge carriers, thereby suppressing non-radiative recombination.

Phase segregation
Reversible or irreversible separation of a mixed-halide perovskite into iodide-rich and bromide-rich domains under illumination or bias, which alters the local bandgap.

Tolerance factor
A dimensionless geometric ratio predicting whether a given combination of A, B and X ions will form a stable perovskite structure.

The value of such systems does not lie in full automation but in improved decision quality. By systematically linking hypothesis generation with experimental validation, closed-loop approaches can reduce reliance on trial and error and help identify more robust design strategies[78]. For PSCs, feedback must also capture variables that strongly affect reproducibility, including processing history, environmental conditions, measurement protocols and device-to-device variability. Progress towards system-level AI is likely to be incremental[38,72,73,85–87] (Table 3): near-term advances will rely on tight integration of local digital and physical loops, ultimately supporting a global framework that continuously propagates knowledge across the PSC research life cycle[88,89].

The future impact of AI in PSC research will depend less on more sophisticated algorithms than on closed-loop AI–experiment integration within adaptive system-level frameworks that preserve the coupled relationships among chemistry, processing, device physics and operational stability. Such a convergence may transform AI from a predictive tool into an active participant in the discovery, optimization and reliability engineering of PSCs.

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Author contributions
Y.L. and G.L. developed the conceptual framework and overall structure of the manuscript, and G.L. supervised the work. Y.L., W.L., Q.C. and G.L. contributed substantially to the discussion of content. Y.L. wrote the article. S.D., M.K.N. and G.L. reviewed and/or edited the manuscript before submission. All authors researched data for the article and contributed to its revision.

Competing interests
The authors declare no competing interests.

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